

Day-1 Progress Report with 3 Real World Examples.

Objectives:-

Today's main objective was to learn about some Security Fundamentals of Cyber Security. So I went with the Video - Learning approach followed by some simple questions. The Topics which I covered today are:

1. CIA TRIAD

Confidentiality: Maintaining Secrecy of data. In simple terms, allowing only authorised users to access data.

Integrity: Maintaining Originality of data. In simple terms, making sure data is not corrupted or altered by an attacker or any other reason.

Availability: Making sure the data is accessible by authorised users at all times.

2. Threats, Vulnerabilities, and Risk

Threats: Anything with potential to harm the system or organisation is a threat.

Vulnerabilities: Any weakness that can be exploited by an attacker is known as a vulnerability.

Risk: The probability of the factors caused by the threats exploiting the vulnerabilities are known as risk. $\text{Risk} = \text{Likelihood} \times \text{Impact}$. Where likelihood is the probability of the risk and impact is the effect on the system or organisation.

3. Attack Surface

Attack Surface: The total area that can be targeted by an attacker is known as the attack surface. In simple terms, the points that the attacker can attack come under the attack surface.

Real World Examples:-

Example 1: Online Banking – CIA Triad

Assuming an net banking website:

- **Confidentiality:** Only the account holder should be able to view their account details.
- **Integrity:** An attacker should not be able to change the account balance or transaction information.
- **Availability:** Customers should be able to access their accounts when they want to make transactions.

This shows the importance of the CIA Triad in the real world.

Example 2: Unpatched Software – Threat, Vulnerability, and Risk

Assuming a company's server is running outdated software containing a known security weakness.

- **Vulnerability:** The outdated software has a known security weakness.
- **Threat:** An attacker who knows about the weakness could attempt to exploit it.
- **Risk:** The company could suffer data loss, service disruption, or unauthorized access if the vulnerability is successfully exploited.

This helped me understand these 3 concepts more clearly.

Example 3: Company Network – Attack Surface

A company may have several systems exposed to potential attacks, such as:

- Public websites
- Internet-accessible servers
- Employee laptops
- Cloud applications
- Email accounts
- Remote-access services

References:-

1. [What is C.I.A Triad](#)
2. [Understanding Threats, Risks, and Vulnerabilities for CC & Security+ Success](#)
3. [What is Attack Surface? Key Concepts for CC and CompTIA Security+ Success 2024](#)